

TANMOY ROY

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SUMMARY

Tanmoy is an AI and Machine Learning (ML) Coach, Research Scholar, Experienced Programmer and Freelance Research Guide. He has diversified experience in the industry as well as in academia and perceived that research is incomplete without practical implementation.

Presently, he guides researchers of **AI and ML** fields and provides training to aspiring AI and ML practitioners. His research area encompasses *speech processing, computer vision, natural language processing, medical imaging, medical data processing and IoT-Edge computing*. As a trainer, he can teach different ML algorithms (*regression, classification, clustering*), advanced topics such as *artificial neural networks, convolutional neural networks*, the latest deep learning models, and cloud deployment of the DL and ML models (*MLOps*). He also teaches programming languages such as *Python, C/C++, R, Java, SQL*.

During PhD, his primary research interest was **speech emotion recognition (SER)**, on which he has few peer-reviewed publications. Notably, he proposed a novel feature set for SER and proposed a novel algorithm for speech endpoint detection. Tanmoy received the *Global Excellence Stature (GES)* scholarship from the **University of Johannesburg (UJ)** to pursue doctoral research and completed his PhD in the stipulated three years. After PhD he joined UJ as a post-doctoral research fellow under the GES fellowship.

Tanmoy started his professional career in **SIEMENS** as a programmer and provided niche solutions to problems in wide-ranging projects encompassing telecommunication, electronic vehicle and smart-grid. His programming skill in different programming languages like C, C++, Java, SQL and expertise in system design flourished during this tenure. He had the opportunity to be part of many complete project development life cycles and learn the craft.

EXPERIENCE

Research and Training Consultancy

KOLKATA, INDIA

[FEB '20 – PRESENT]

Research Consultant

As a research consultant got the opportunity to handle diversified research projects. Research topics encompass computer vision (CV), Natural Language Processing (NLP), Data Science (DS) and more. The projects are listed below:

Human Fall Detection (CV): Working on a *Human Fall Detection system*. In this project, the frame images of surveillance video cameras are used to detect the falling person. Latest *Deep Learning (DL)* techniques are being evaluated for different modules. Proposed a *Masked R-CNN based semantic segmentation model* [2] which achieved an f1-score of 94% and IoU of 89%. The next sub-module to develop is a *human pose estimation model*.

Face Mask Detection (CV): Worked on a *mask detection system* to detect face masks from surveillance cameras during the Covid-19 pandemic. Used *MobileNet V2* and *Inception ResNetV2* transfer learning model.

Text Classification (NLP): Developed a *Multi-head Attention* based text classification model [1] and achieved more than 96% test accuracy. Compared the results with other prominent classification models such as *naive bayes, logistic regression, random forest and multi-layer perceptron*. Looking forward to enhancing this model.

Polyp Detection (Medical Imaging): Working on a U-Net based architecture to detect polyps in colonoscopy images. This model will be helpful in detecting cancerous polyps.

Electronic Health Record (EHR) and Process Mining: This is a unique project where the problem is to predict the possibility of re-admission of a patient within 30-days of discharge based on the available EHR. Here the process mining concepts are used to extract features from the EHR and then deep learning sequence-to-sequence models are trained to predict re-admission possibility. Test accuracy of the model is 88% which is encouraging keeping in view of the system complexity and novelty of the work.

IoT-Edge Computing: Prepared a PhD research proposal on an Edge computing based IoT forensic model. The proposed model consisted of modules such as *online incident identification, evidence identification and management, live threat detection and forensic data reduction logic*. The modules are primarily based on deep learning architecture.

Training Consultant

As a trainer, he guides aspiring AI&ML practitioners to grasp critical concepts. Can offer different training tracks to people from various backgrounds such as students, people switching careers, people starting fresh, and people upgrading skills. He is training on a wide range of topics listed as follows:

Data Science and Machine Learning Concepts and algorithms of data visualization, statistical analysis, regression analysis, classification, clustering and time-series analysis. Provides in-depth explanation as and when required supported by suitable hands-on examples using python and R. Can explain topics at different depths depending on the audience.

Deep Learning (DL): Theory and implementation of artificial neural networks, convolutional neural networks and recurrent neural networks to the latest *attention* based models. Keeps track of the latest DL models being published.

Python for ML and DL: Different python libraries such as *numpy, scipy, pandas, pytorch, tensorflow* as ML and DL tools. Python basics, object oriented concepts and general purpose utilities such as web programming (*flask*) for ML and DL model deployment.

Practical Machine Learning (MLOps): Building MLOPs pipeline from Packaging ML and DL models, Continuous Integration, Continuous Delivery, Publishing using web interfaces (e.g.*flask*) and finally Maintenance and Reporting.

Cloud and Docker container: AWS data toolchain from acquisition (e.g. DMS), storage (e.g. RDS), processing (e.g. EMR) and presentation (SageMaker) or modelling (Deep Learning). Creating and Running ML & DL Docker containers.

University of Johannesburg

JOHANNESBURG, SOUTH AFRICA

[JUL '16 – SEP '21]

Post Doctoral Research Fellow

[OCT '19 – SEP '21]

Built a *feed-forward neural network* (FNN) model [3] using the feature set proposed [8] in his doctoral studies. Written two survey papers, one highlighting the role of emotion recognition in the context of the 4th-industrial revolution [5] and the other one summarizing the classification techniques in SER [4]. During this tenure, he initiated two *research collaborations* one with King Saud University [2] in Riyadh, Saudi Arabia and Birla Institute of Technology and Science [1] in Pilani, India.

Doctoral Research Fellow

[JUL '16 – JUN '19]

Completed doctoral study on his thesis titled "Detecting emotions from speech using Machine Learning Techniques" within stipulated three year time. Proposed a novel feature set for SER named SEGRT(*subjective emotional gap reduction technique*) [8, 6]. Also proposed a new algorithm for precise detection of speech endpoints to enhance the pre-processing of speech signals [7].

SIEMENS

KOLKATA, INDIA

[Jan '07 – Sep '15]

Associate Consultant

[Jan '13 – Sep '15]

- Designed and implemented a Telecom Service Provisioning system based on a RedHat Linux multi-tiered application using Java Spring Framework. Technologies used *Java Spring, Oracle*.
- Customization of Telecom Service Provisioning system using *Java, SOAP, XML, Oracle* on *Linux* platform.
- Designed and implemented CRM Service Provisioning integration module for GSM-R Billing system using *Java, Oracle PL/SQL*.
- Designed and implemented the Billing module for GSM-R Billing system using *C/C++, Oracle PL/SQL* on *Linux* platform.

Sr. System Engineer

[Jan '11 – Dec '12]

- Designed and implemented a Charging module for the Electronic Car billing system using *C/C++ and Oracle*.
- Designed and implemented GUI based Telecom Call Analysis system. Features include Call tracking, Zone based call analysis with Graphical simulations. Used *C++ based Qt GUI Library*.

System Engineer

[Jan '09 – Dec '10]

- Designed and implemented Bill Printing and Bill By Email module for a Telecom Billing system. **Technologies:** Java, ProC/C++, Oracle PL/SQL.
- Designed and implemented a Charging module for a Telecom Interconnect Billing system using Java and Oracle. **Technologies:** Java, Oracle.

Associate System Engineer

[Jan '07 – Dec '08]

- Single handedly provided System Support for IVR and USSD Solution. Technologies used: *Java, C++, Linux, awk script, shell script*.
- Developed an automated monitoring system using *Java, shell script, awk script* to improve client experience.

RESEARCH EXPERTISE

- Tanmoy got attracted to research during his professional career in SIEMENS and was involved in many research oriented works. Some of his solutions are still in use.
- During his PhD he proposed a novel feature set for SER [8, 6] and a novel algorithm for speech endpoint detection which is published in a high impact journal [7].
- He is interested in inter-disciplinary research where ML&DL techniques can be applied to solve problems in other domains. He is presently working on computer vision [2], natural language processing [1] and medical imaging research where the latest DL models are being used.
- So far he has 8 publications with reputed publishing houses and looking forward to publishing more in individual capacity and in collaboration as well.

RESEARCH SUPERVISION & GUIDANCE

- **Guiding a PhD scholar from King Saud University, Saudi Arabia.**
Area of Research: Computer Vision
Research Title: Detection of human fall from surveillance camera.
- **Supervising an MTech student from BITS Pilani, India.**
Area of Research: Computer Vision
Research Title: Visual Scene Understanding for Autonomous Vehicle.

TEACHING EXPERTISE

Machine Learning and Data Science topics:

- Data scaling: normalize, standard scaler, one-hot.
- Data cleaning: missing value imputation, outlier detection.
- Feature selection: feature importance, correlation, Covariance, t-test, F-test, χ^2 test, ANOVA.
- Feature extraction: PCA, LDA, ICA, SVD, NMF.
- Hypothesis testing, parametric and non-parametric tests.
- Regression: OLS, Ridge, Lasso, Elastic Net, Polynomial, Bayesian Ridge, SGD, Gaussian Process.
- Classification: Logistic Regression, KNN, Naïve Bayes, Decision Tree, Ensemble Methods- Bagging (Random Forest), Boosting (Ada Boost, Gradient Boosting, XGBoost).
- Clustering: KMeans, Hierarchical

Deep Learning models: ANN, CNN, RNN, LSTM, Auto-Encoders, Attention

Programming: Python, C/C++, SQL, R, Java.

TECHNICAL EXPERTISE

- **Programming Languages:** *Python, C/C++, R, Matlab, Java, JavaScript, SQL, Scala, Julia.*
 - **Libraries:** Few prominent python libraries are: *pytorch, tensorflow, keras, theano, numpy, pandas, matplotlib, sns, scipy, opencv, nltk, flask, pyspark.*
 - **Technologies:** Cloud (AWS, GCP), Linux, Docker container, MLOps, REST, SOAP, XML, Gradle Build.
 - **Databases:** RDBMS (*Oracle, PostgreSQL, SQL Server, MySQL*), NoSQL (*MongoDB, DynamoDB, Cassandra*), Cloud DB (*Amazon RDS*),
 - **Data Visualization:** Tableau
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EDUCATION

Doctor of Philosophy

2019, UNIVERSITY OF JOHANNESBURG

JOHANNESBURG, SOUTH AFRICA

Thesis Title: Detecting Emotions From Speech Using Machine Learning Techniques.

Area of research: Applied Machine Learning and Deep Learning, Speech Signal Processing.

Master of Science

2013, LOVELY PROFESSIONAL UNIVERSITY

KOLKATA, INDIA

Specialization: Information Technology.

Subjects: Computer Organization, Programming(C++/Java), Data Structure, Computer Graphics, Artificial Intelligence, System Software.

Post-Graduate Diploma in Computer Applications

2004, DOEACC SOCIETY, MINISTRY OF IT, GOVT. OF INDIA.

KOLKATA, INDIA

Advanced Diploma in Computer Applications (A-Level).

Bachelor of Science

2002, UNIVERSITY OF CALCUTTA

KOLKATA, INDIA

Economics(Hons), Computer Science, Mathematics

CERTIFICATES

1. **Deep Learning with PyTorch : Neural Style Transfer**
Coursera
 2. **Neural Network from Scratch in TensorFlow**
Coursera
 3. **Billing and Revenue Management**
Oracle University
 4. **Diploma in Computer Networking**
CMC Limited
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PUBLICATIONS

- [1] Sinha J, Roy T, and Marwala T. "Attention Based Encoder Architecture for Automatic Text Classification: A Case Study on Text-news Classification". In: *Mathematical Methods in Dynamical Systems*. Ed. by S Chakraverty and S Jena. CRC Press (Taylor & Francis Group), 2021.
- [2] Alanazi T, Roy T, and Marwala T. "Generating Human Segmentation Dynamically in Multi-camera Setup Using Deep Learning". In: *Mathematical Methods in Dynamical Systems*. Ed. by S Chakraverty and S Jena. CRC Press (Taylor & Francis Group), 2021.
- [3] Roy T, Marwala T, and Chakraverty S. "Chapter 12 - Speech emotion recognition using deep learning". In: *New Paradigms in Computational Modeling and Its Applications*. Ed. by Snehashish Chakraverty. Academic Press, 2021, pp. 177–187. ISBN: 978-0-12-822133-4. DOI: <https://doi.org/10.1016/B978-0-12-822133-4.00009-8>.

- [4] Roy T, Marwala T, and Chakraverty S. "A Survey of Classification Techniques in Speech Emotion Recognition". In: *Mathematical Methods in Interdisciplinary Sciences*. John Wiley & Sons, Ltd, 2020. Chap. 3, pp. 33–48. ISBN: 9781119585640. DOI: <https://doi.org/10.1002/9781119585640.ch3>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781119585640.ch3>.
 - [5] Roy T, Marwala T, and Chakraverty S. "Advancements and Role of Emotion Recognition in the 4th Industrial Revolution". In: *The Disruptive Fourth Industrial Revolution: Technology, Society and Beyond*. Ed. by Wesley Doorsamy, Babu Sena Paul, and Tshilidzi Marwala. Cham: Springer International Publishing, 2020, pp. 179–203. ISBN: 978-3-030-48230-5. DOI: [10.1007/978-3-030-48230-5_8](https://doi.org/10.1007/978-3-030-48230-5_8).
 - [6] Roy T, Marwala T, and Chakraverty S. "Speech Emotion Recognition Using Neural Network and Wavelet Features". In: *Recent Trends in Wave Mechanics and Vibrations*. Ed. by S Chakraverty and Paritosh Biswas. Singapore: Springer Singapore, 2020, pp. 427–438. ISBN: 978-981-15-0287-3. DOI: [10.1007/978-981-15-0287-3_30](https://doi.org/10.1007/978-981-15-0287-3_30).
 - [7] Roy T, Chakraverty S, and Marwala T. "Precise detection of speech endpoints dynamically: A wavelet convolution based approach". In: *Communications in Nonlinear Science and Numerical Simulation* 67 (2019), pp. 162–175. ISSN: 1007-5704. DOI: <https://doi.org/10.1016/j.cnsns.2018.07.008>.
 - [8] Roy T et al. "Introducing New Feature Set based on Wavelets for Speech Emotion Classification". In: *2018 IEEE Applied Signal Processing Conference (ASPCON)*. 2018, pp. 124–128. DOI: [10.1109/ASPCON.2018.8748666](https://doi.org/10.1109/ASPCON.2018.8748666).
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SCHOLARSHIPS

- Received Global Excellence Stature (GES) Doctoral fellowship at the University of Johannesburg in 2016.
 - Received GES Post Doctoral Fellowship at the University of Johannesburg in 2019.
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JOURNAL REVIEWER

- **International Journal of Systems Science.**
Tailor and Francis
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REFERENCES

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